Figure 3.8 According to general relativity, not only will the spacetime loaf be sliced into space at moments of time at different angles (by observers in relative motion), but the slices themselves will be warped or curved by the presence of matter or energy.

Figures 3.9, 3.10, and 3.11 schematically illustrate the triumph of Einstein's ten-year struggle. Much of his work during these years aimed at determining the precise shape and size of the warping that would be caused by a given amount of matter or energy. The mathematical result Einstein found underlies these figures and is embodied in what are called the *Einstein field equations.* As the name indicates, Einstein viewed the warping of spacetime as the manifestation—the geometrical embodiment—of a gravitational field. By framing the problem geometrically,

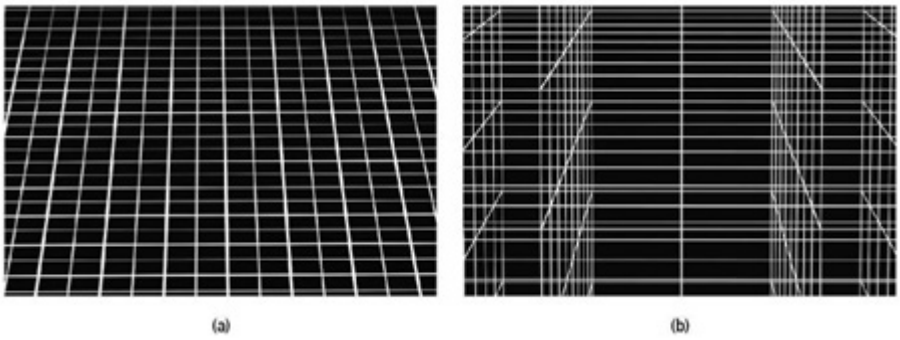


Figure 3.9 **(**a**)** Flat space (2-d version). **(**b**)** Flat space (3-d version).

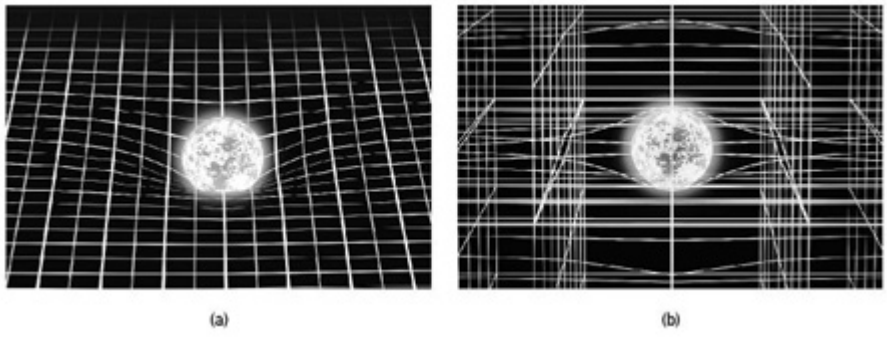


Figure 3.10 **(**a**)** The sun warping space (2-d version). **(**b**)** The sun warping space (3-d version).

Einstein was able to find equations that do for gravity what Maxwell's equations did for electromagnetism.[16] And by using these equations, Einstein and many others made predictions for the path that would be followed by this or that planet, or even by light emitted by a distant star,